

Seven-Color Spotlight Alternating

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[Feature Description](#)

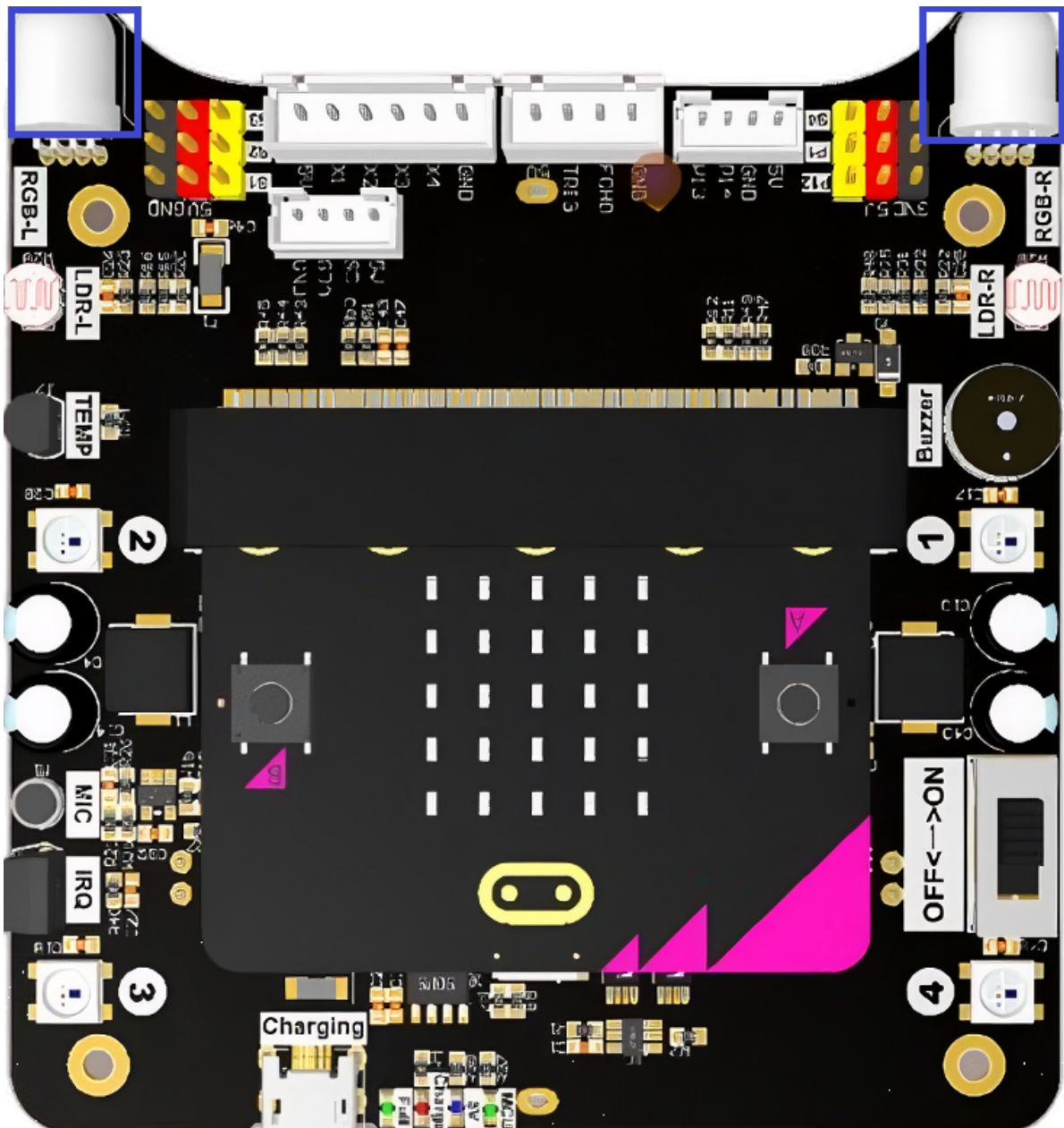
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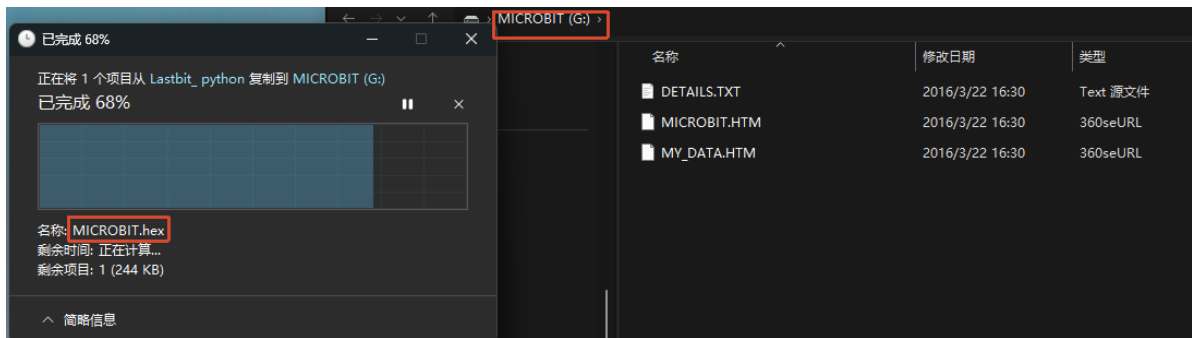
Feature Description

The front of the Lastbit car is equipped with two RGB spotlights, RGB_L and RGB_R. In this section, we will learn how to control the RGB lights to light up alternately, presenting a flowing light effect.



Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

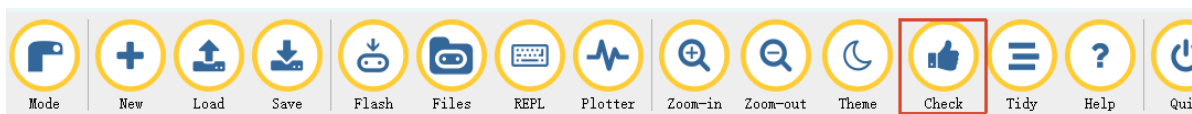
Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

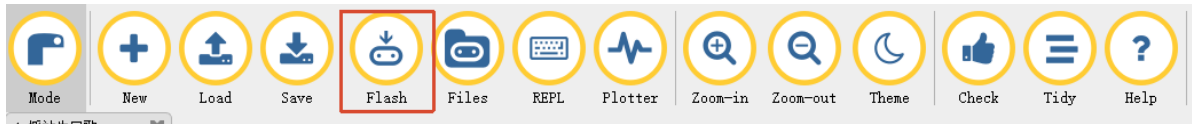
4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.



6. Switch the switch to the ON position.

Code Implementation

```
# 2.RGB_Spotlight_Alternating

# Alternate RGB searchlight illumination
# 2 seconds per switching state
#
# on all lights 2 seconds
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# close all lights
Lastbit.all_lights_off()

while True:
    # left red right close
    Lastbit.set_headlight_left_right('RED', 'CLOSE')
    sleep(2000)

    # right red left close
    Lastbit.set_headlight_left_right('CLOSE', 'RED')
    sleep(2000)

    # left yellow right close
    Lastbit.set_headlight_left_right('YELLOW', 'CLOSE')
    sleep(2000)

    # right yellow left close
    Lastbit.set_headlight_left_right('CLOSE', 'YELLOW')
    sleep(2000)

    # left purple right close
    Lastbit.set_headlight_left_right('PURPLE', 'CLOSE')
    sleep(2000)

    # right purple left close
    Lastbit.set_headlight_left_right('CLOSE', 'PURPLE')
    sleep(2000)
```

The function methods have already been introduced in the previous tutorial, so they will not be repeated here.

Experimental Phenomenon

The car spotlights will present the following sequence: left red and right off, right red and left off, left yellow and right off, right yellow and left off, left purple and right off, right purple and left off. The effect switches every 2 seconds and runs in a loop.